

3.4 Unit 4 PSYB4 Approaches, Debates and Methods in Psychology

Students should have experience of designing and conducting informal classroom research using a variety of methods. They will be expected to analyse data collected in investigations and draw conclusions

based on research findings. They will be required to draw on these experiences to answer questions in the examination for this unit.

Approaches in Psychology

Topics

3.4.1 Biological, Behaviourist, Social Learning, Cognitive, Psychodynamic and Humanistic Approaches

3.4.2 Comparison of Approaches

Aims

- To enable the development of understanding and critical appreciation of the breadth of theoretical and methodological approaches in psychology.
- To enable students to build on knowledge, understanding and skills developed in other units.
- To encourage appreciation of the contrasting and complementary aspects of different approaches in psychology.
- To develop an appreciation of how science works in psychology.

3.4.1 Biological, Behaviourist, Social Learning, Cognitive, Psychodynamic and Humanistic Approaches

Biological Approach

Assumptions and application of the biological approach. The role of the central and autonomic nervous system in behaviour. The genetic basis of behaviour. Strengths and limitations of the biological approach.

Behaviourist Approach

Assumptions and application of the behaviourist approach. Key concepts including stimulus, response and reinforcement. Types of reinforcement. Classical and operant conditioning as applied to human behaviour. Strengths and limitations of the behaviourist approach.

Social Learning Theory

Social learning theory as a bridge between traditional behaviourism and the cognitive approach.

Cognitive Approach

Assumptions and application of social learning theory. The role of mediational processes in learning, motivation and performance of behaviour. Observational learning and the role of vicarious reinforcement. Strengths and limitations of social learning theory

Assumptions and application of the cognitive approach, including the idea that thoughts influence behaviour. Information processing and how this applies to human behaviour and thought. Use of computer analogies in understanding behaviour. Strengths and limitations of the cognitive approach.

Psychodynamic Approach

Assumptions and application of the psychodynamic approach. Freud's approach to personality structure and dynamics. Unconscious mental processes. Psychosexual stages of development. Freud's use of case studies to highlight concepts. Post-Freudian theories including Erikson and at least one other. Strengths and limitations of the psychodynamic approach.

Humanistic Approach

Assumptions and application of the humanistic approach. The person-centred approach of Rogers and Maslow. Rejection of the traditional scientific approach and experimentation. The importance within humanistic psychology of valuing individual experience, promoting personal growth, the concepts of freewill and holism. Strengths and limitations of the humanistic approach.

3.4.2 Comparison of Approaches

Comparison of biological, behaviourist, social learning, cognitive, psychodynamic and humanistic approaches. The extent to which different approaches overlap and complement each other. The value of individual approaches and the merits of taking an eclectic approach to explaining human behaviour and in the application of psychology.

Debates in Psychology

Topics

Aims

3.4.3 Debates in Psychology

- To enable students to engage in key critical debates in psychology.
- To enable students to build on knowledge, understanding and skills developed in other units.
- To give students a critical understanding of the features and principles of the scientific approach.
- To enable students to relate the different approaches in psychology to the nature of scientific enquiry and the key critical debates.
- To develop an appreciation of debates relating to how science works in psychology.

3.4.3 Debates in Psychology

Free will and determinism. Hard determinism and soft determinism. Biological, environmental and psychic determinism. The scientific emphasis on causal explanations.

The nature-nurture debate. The relative importance of heredity and environment in determining behaviour. An interactionist approach.

Holism and reductionism. The strengths and limitations of reductionist and holistic explanations. An interactionist approach.

Idiographic and nomothetic approaches. The strengths and limitations of idiographic versus nomothetic research.

Psychology and science. The features and principles of the scientific approach: a paradigm; the role of theory; hypothesis testing; empirical methods and replication; generalisation. The subject matter of psychology: overt behaviour versus subjective, private experience. The role of peer review in validating research.

Strengths and limitations of the scientific approach in psychology.

Methods in Psychology

Topics

Aims

3.4.4 Inferential Statistics

3.4.5 Issues in Research

- To build on knowledge of research methods and statistics acquired in previous units.
- To enable understanding of the concepts of hypothesis testing and significance.
- To promote an understanding of significance testing using inferential statistics.
- To inform decision making about the appropriateness of different inferential tests in different circumstances.
- To enable the development of a critical understanding of research issues in psychology.
- To develop an appreciation of how science works in psychological research.

3.4.4 Inferential Statistics

Statistical inference

The concepts of probability and levels of significance. Hypothesis testing: null and alternative (experimental or research) hypothesis. One and two tailed tests. Type I and Type II errors. Positive, negative and zero correlation. Limitations of sampling techniques and generalisation of results.

Statistical tests

Use of non-parametric and parametric tests. Statistical tests of difference: the sign test; Wilcoxon signed ranks test; Mann-Whitney; related t-test (repeated measures) and independent t-test. Statistical tests of association: Spearman's rank order correlation; Pearson's product moment correlation; the Chi-square test.

Factors affecting the choice of statistical test, including levels of measurement, type of experimental design. Criteria for parametric testing: interval data; normal distribution; homogeneity of variance.

3.4.5 Issues in Research

Strengths and limitations of different methods of research. Strengths and limitations of qualitative and quantitative data.

Reliability and validity applied generally across all methods of investigation. Types of reliability and validity. Ways of assessing reliability and validity.

Critical understanding of the importance of ethical considerations within the social and cultural environment. Ethical considerations in the design and conduct of psychological studies and within psychology as a whole.
